Printed Pages:2 **Sub Code:KCA-052**

Paper Id: 236765 **Roll No.** | | | | | | | | | | |

MCA
(SEM IV) THEORY EXAMINATION 2022-23
COMPUTER GRAPHICS AND ANIMATION

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

a. Define Random Scan/Raster Scan displays.
b. What is Aspect Ratio?
c. What is Scaling?
d. Distinguish between window and viewport.
e. What is view plane normal vector?
f. What is the use of control points?
g. What are the types of reflection of incident light?
h. Define shading.
i. Define frame.
j. Define complementary colors.

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 30**

a. Calculate the pixel location approximating the first octant of a circle having centre at (4,5) and radius 4 units using Bresenham's algorithm.

b. Explain the following composite transformations :
(i) Traslation (ii) Rotation

c. With suitable examples explain all 3D transformations.

d. What is illumination? What are the various types of illumination.

e. What is tweening? Explain in detail about motion tween with an example.

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

a. Explain in detail about the Line drawing DDA scan conversion algorithm?

b. Explain refresh CRT and color CRT monitors.

4. Attempt any *one* part of the following: **10 x 1 = 10**

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a. Explain in detail about two dimensional viewing.

b. Write about Cohen-Sutherland line clipping algorithm with an example.

5. Attempt any *one* part of the following: 10 x 1 = 10

a. Write short notes on Bezier curve and spline.

b. Discuss any one of the visible surface detection method.

6. Attempt any *one* part of the following: 10 x 1 = 10

a. Explain in detail the CMY color model.

b. Discuss in detail Gouraud shading.

7. Attempt any *one* part of the following: 10 x 1 = 10

a. What is Animation? Describe the various animation techniques.

b. What is storyboarding? Discuss in detail.

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